Efficiency Analysis

Detection of inefficient operators for CallMeMaybe through data analytics and statistical scoring.

What does this analysis cover?

Missed Calls

Identification of high abandonment rates in incoming calls by operators.

Wait Times

Measurement of excessive queue latency that directly impacts customer satisfaction.

Low Activity

Analysis of the volume of outgoing calls as an indicator of work proactivity.

Evaluation

Generation of a unified score to guarantee the quality of requirements compliance.

Scoring Algorithm

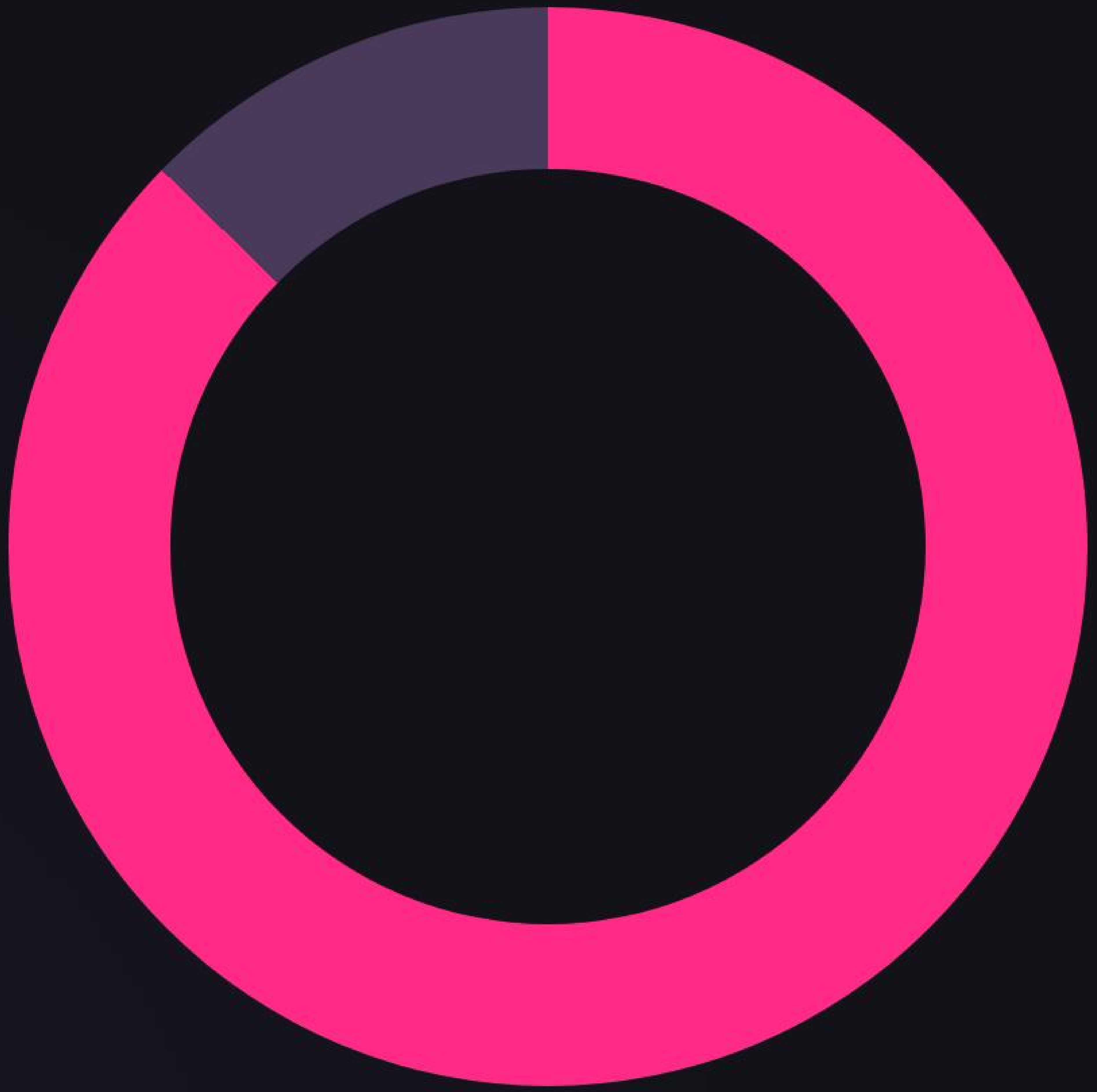
Weighted Min-Max normalization formula:

$$\text{Score} = 0.5 \cdot T_{\text{missed}} + 0.4 \cdot \text{Wait} + 0.1 \cdot \text{Inactivity}$$

The model prioritizes customer impact (90% weight on wait and missed calls), allowing the isolation of the lowest-performing 95th percentile automatically.

Traffic Distribution

Call Volume



External: 39,861 (87.3%)

Internal: 5,809 (12.7%)

The system is a fundamental pillar for external customer service, which validates the need to optimize every interaction.

Directional Traffic

The operation is predominantly proactive. A drop in outgoing calls is a critical indicator of inefficiency if there is no increase in support.

Statistical Validation

$p < .05$

SIGNIFICANCE

T-test Verifications

The t-test confirms significant differences in wait times.

A positive correlation was validated between wait time and the abandonment rate.

Strategic Action Plan

Training

Agility training for operators in the 95th percentile of inefficiency.

Dashboard

Implementation of real-time alerts for wait time peaks in the system.

Balancing

Dynamic redistribution of workload based on historical call flows.

Iteration

Periodic adjustment of score weights according to business KPI evolution.

Stack and Documentation

TOOL	STRATEGIC APPLICATION
Pandas	Null handling, cleaning, and transformation of time series.
Dash / Plotly	Reactive dashboard development and proactive data visualization.
SciPy Stats	Hypothesis validation and calculation of statistical significance.
Scikit-Learn	Metrics normalization for composite score generation.